DataBaseAgent: a local desktop workbench for building topic-specific biological data repositories with human-gated AI agents

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Draft of 5 November 2026

Abstract

Agent-curated repositories such as scBaseCount show that large language model (LLM) agents can discover, annotate and reprocess public sequencing data at a scale manual curation cannot reach. That demonstration is a single bespoke repository: its agent, pipeline and cloud deployment were engineered for one assay by one institute. Most researchers need something smaller and more specific, such as every public dataset on their organism, tissue or perturbation of interest, processed uniformly with tools they trust, and few have the engineering capacity to build it; GPSAdb, a manually curated compendium of 3,048 human gene-perturbation RNA-seq datasets, is the authors' own instance of that need. Here we present DataBaseAgent (DBA), a general-purpose desktop workbench that turns the agent-curated repository pattern into a tool any researcher can run. DBA ships as one binary for Windows, macOS and Linux with no server or database to deploy. A project states its topic, organism, assay and inclusion criteria in free text. Candidates are gathered from Europe PMC, PubMed or GEO, or imported. An LLM reviewer proposes include, exclude or uncertain with a verbatim evidence snippet that the application verifies against the record; a dataset enters the repository only through a recorded decision that carries a written reason and names who made it. The user writes the processing pipeline as ordinary shell steps and locks a version. Jobs run on the user's own Linux server over SSH under CPU and concurrency quotas; network failures are retried a bounded number of times and everything else is escalated to the user with the log and a resume point. Outputs are checksummed, manifested and archived locally. One project agent and four specialists, each with its own model, instructions and fixed tool list, work in a single conversation per project; they act only through the same validated commands as the interface, run only when addressed, and every call is recorded in the conversation. Any OpenAI-compatible, Anthropic or Gemini endpoint, including locally served models, can be the model. We present DBA as a design contribution: its workflow and architecture, the safeguards around acting agents, two walkthroughs on live services, one of which carries a single GPSAdb-style dataset from a GEO search through a help request to a checksummed count matrix, and a feature comparison with repository-scale curation agents, dataset fetchers, screening tools and workflow platforms. Usability and the quality of the repositories users build with it remain to be evaluated. DBA is open source at [repository URL].

Keywords: data curation, AI agents, human-in-the-loop, reproducible pipelines, desktop software, sequencing data repositories

Preprint status: draft for bioRxiv (Bioinformatics). Software name in the text: DataBaseAgent (DBA). Comparator named in the brief as "ScBaseAgent" is scBaseCount with its SRAgent curation agent (Youngblut et al., Cell 2026); the draft uses the published names.

Draft conventions: square brackets mark placeholders the authors must fill or confirm; every quantitative claim about DBA comes from the repository (README, tests, acceptance record and single-

dataset case record of 5 November 2026, at commit 37104a6) and is cited to it; nothing about runtime performance or user studies is claimed because none has been measured.

1 Introduction

Public sequencing archives hold far more data than any community uses. The Sequence Read Archive (SRA) and the Gene Expression Omnibus (GEO) grow faster than curated atlases can absorb them, and most of their content is reachable only through free-text metadata that was written for submission, not for reuse [2,3]. Uniformly reprocessed resources such as ReCount and recount3, ARCHS4 and refine.bio showed that re-running one pipeline over thousands of public experiments removes processing artefacts and makes cross-study analysis routine [4,5,6]. Each of them, however, is a centrally built product with a fixed scope, and the labour of selecting, annotating and reprocessing datasets is paid once by one team.

scBaseCount moved that labour to an AI agent [1]. Its curation agent, SRAgent, is a hierarchy of ReAct-style language-model agents [8] that queries NCBI tools, decides whether an SRA experiment is 10x Genomics single-cell data, extracts organism, tissue and disease, and stores the result in a relational database; a Nextflow pipeline, scRecounter, then reprocesses every accepted experiment [9]. The outcome is the largest public single-cell repository, more than 502 million cells from 27 organisms, and the authors show that agentic extraction recovers metadata that deterministic tools such as ffq and pysradb cannot [1,16,17]. They also state the general lesson: the approach "provides a blueprint for how AI can be leveraged to autonomously curate biological data repositories" [1]. The blueprint itself, though, is not something a typical laboratory can pick up. SRAgent runs on Google Cloud Run against a hosted PostgreSQL database, scRecounter runs on GCP Batch, the pipeline is specific to 10x chemistry, and the published run consumed an estimated 13.2 trillion tokens at a cost of about 17,000 USD [1].

Most repository needs sit in the long tail below that scale, and the authors' own database is the motivating case. GPSAdb collects human cell-line RNA-seq datasets in which one gene was knocked out or down: 3,048 datasets for 1,458 genes in its published version, each found by searching GEO and ENCODE with knockdown and knockout terms, screened by hand, given its control and perturbed samples by hand, and reprocessed with one pipeline into differential expression tables and 6,096 gene sets that its GPSA tool uses to infer candidate causal perturbations from a user's own expression data [32]. The paper promised an annual survey of new data; the database registry lists its last update as August 2022 [33], because every update repeats the whole manual cycle. Other groups face the same cycle: a plant group wants every public single-cell dataset of a root tissue, a disease group every bulk cohort that matches a phenotype, a perturbation group every screen that targeted a pathway. Today such a group stitches together a search in PubMed or Europe PMC, a spreadsheet of candidates, a fetching tool such as pysradb, ffq or nf-core/fetchngs [15,16,17], a workflow manager [9,10,25], and a cluster account, and then keeps the whole arrangement alive by hand. Platforms such as Galaxy and Seqera cover execution well but start after the datasets have been chosen [11]; screening tools such as ASReview and Rayyan triage literature but stop before any data is touched [14,23]; and general-purpose biomedical agents such as Biomni and CellAgent analyse data already in hand without addressing which datasets to gather [13,22]. Building the repository is hard. Maintaining it is harder: new datasets keep appearing, the pipeline changes, remote jobs fail at night, and the reasons a dataset was included are lost in a chat log or a colleague's memory.

Here we present DataBaseAgent (DBA), a local desktop workbench that packages the agent-curated repository pattern as a general tool. It grew out of the need to keep GPSAdb current, and the sample groups it records beside each dataset and the shape of its manifests come from that database's needs, but nothing in the tool is specific to it. Seven design commitments shape it. First, generality by construction: topic, organism, assay and inclusion criteria are free text,

the pipeline is whatever shell commands the user would have typed, and the output contract is a list of relative file paths. Second, gated decisions: a reviewer agent must quote verbatim evidence from the record and the application checks the quotation; a dataset is included, a pipeline locked or a failed job resumed only by an explicit, recorded decision that carries a reason and its author, whether a person issued it or an agent did so on the person's instruction. Third, sparse agents: no agent process is resident. Agents are invoked at decision points, when a reviewer is asked for a suggestion, when a user states a goal, when a failed job is sent for diagnosis, and the only long-running work is the user's own pipeline on the user's own server. A light scheduling tick reads state, dispatches what quotas allow, and goes back to sleep. Fourth, separation of orchestration from computation: the desktop keeps the state and the decisions, the server keeps the raw data and the compute, and a closed laptop does not stop a running job. Fifth, least privilege and traceability: each agent has a fixed, project-scoped tool list, its commands pass through the same validation as the interface's forms, tool output is treated as untrusted text, and each call with its arguments, result and latency is written into the conversation. Sixth, maintainability as a first-class goal: pipeline versions are locked and snapshotted into each job, submission is idempotent across lost connections, searches merge and deduplicate into the existing candidate list, every result carries a manifest and a SHA-256, and a restart resumes monitoring where it stopped. Seventh, zero deployment: DBA is one binary that stores a JSON file, keeps model keys in the operating system's credential store, talks to models over three common protocols, and reaches servers with the system SSH client.

We present DBA as a design contribution in the sense used by Rho [12]: Figure 1 states the argument, and the paper describes the workflow, the architecture, the safeguards and two walkthroughs, one on live search services and one that carries a single dataset end to end, and it compares DBA feature by feature with repository-scale curation agents, dataset fetchers, screening tools and workflow platforms. It does not claim that repositories built with DBA are better than hand-built ones, or that the tool is usable without training; both remain to be evaluated. The contributions are: (i) a general, local-first realisation of the discover, review, process and archive loop that scBaseCount performed for one assay; (ii) a sparse, least-privilege multi-agent design in which agents act only through validated, attributed commands and every action is auditable; (iii) a scheduling and recovery model for unattended remote processing with bounded retries and explicit hand-over to a human; and (iv) an open-source implementation with an automated test suite covering project isolation, review gating, quota enforcement, recovery, archival, tool validation and delegation.

2 Results

2.1 The DBA workflow: six stages and three gated decisions

DBA organises work as projects, and a project is one repository under construction. It holds a topic, an organism, an assay, inclusion criteria written as prose, the search expressions tried and their sources, the candidate list with its review record, a pipeline draft with its locked versions, and one conversation shared by the project agent and the specialists it calls. Several projects coexist in one workspace and share the configured servers and models, while quotas are enforced per server and per project. Figure 2 shows the six stages; three of them end in a decision the application never takes on its own.

1. **Define.** The user writes the topic and the inclusion criteria in plain language. There is no schema to design: the same criteria text is what the reviewer agent and the human reviewer read.
2. **Search.** A deliberately broad query goes to Europe PMC, PubMed or GEO. DBA pages through results up to a user-set cap, records the hit count, the number fetched and the

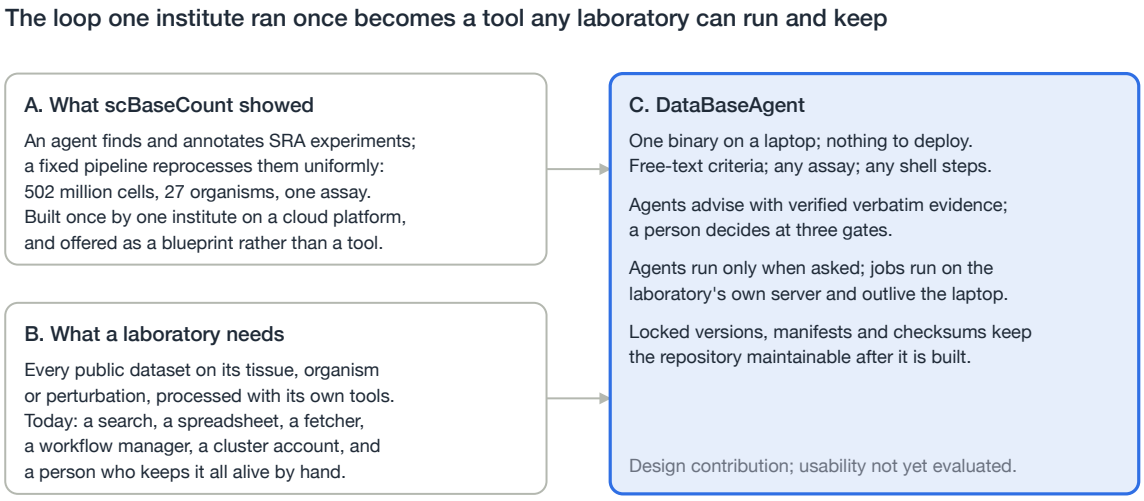


Figure 1: **The argument.** (A) scBaseCount showed that an agent can discover, annotate and feed a uniform pipeline at repository scale, as a one-off build by one institute [1]. (B) A laboratory's need is narrower and recurring, and is met today by a chain of tools and a person; GPSAdb is the authors' instance of it [32]. (C) DBA is the same loop as a tool: general by construction, gated by recorded decisions, sparse in its use of agents, run on the laboratory's own server, and designed to be maintained.

number newly added, and merges records into the existing candidate list by normalised title, DOI and accession, so a repeated or refined search adds only what is new. Candidate lists can also be imported as JSON from any other source, which is how ENA, ArrayExpress or a colleague's spreadsheet enter a project.

3. **Review (gate 1).** For each pending candidate the reviewer agent returns include, exclude or uncertain, a confidence level, a reason and an evidence string. DBA demotes any suggestion whose evidence is not a verbatim substring of the title or abstract to uncertain with low confidence and a note asking for manual inspection. The candidate is then marked accepted or excluded by a decision that must carry a written reason and records its author, the user or an agent acting on the user's instruction in the conversation, and may add the accession, a direct download address and sample groups. Only accepted candidates can be queued.
4. **Pipeline (gate 2).** The user writes the processing steps as shell commands that use six environment variables (accession, source URL, input directory, output directory, sample groups, CPU count), lists the relative paths of the expected outputs, chooses a server, a CPU count and a per-project concurrency, and locks the version. Each job snapshots the locked pipeline, so a later edit creates a new version and never alters a running or finished job.
5. **Run.** A scheduler tick, by default every few seconds and configurable, promotes queued jobs whose server CPU quota, server job limit and project concurrency allow, submits them over SSH with an idempotent marker, and polls their exit code and process id. Transient network failures are retried with exponential back-off up to a configured limit; any other failure, and any state that cannot be determined, moves the job to a needs-help state with the tail of the remote log and a recorded resume point. **Gate 3:** the person reads the diagnosis, optionally produced by the operations inspector agent, fixes the environment or supplies the missing information, and resumes the job from the job dialog or by instructing the operations inspector, which resumes it with the same validated command.
6. **Archive.** Each declared output is hashed on the server, streamed to the workspace with

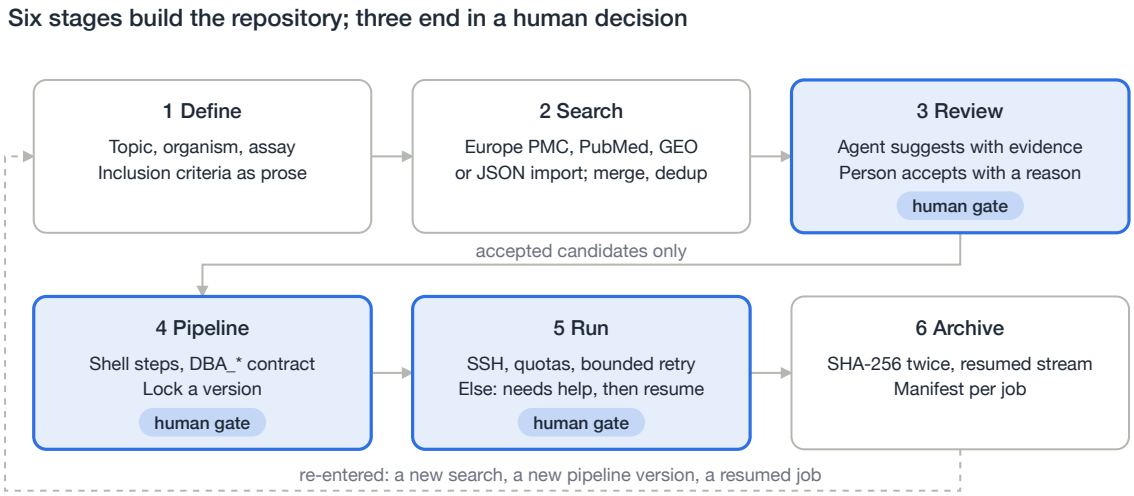


Figure 2: **The DBA workflow.** Six stages take a project from a topic statement to an archived, manifested result. The three tinted stages end in a decision the application never takes on its own: accepting a candidate with a written reason, locking a pipeline version, and resuming a job the scheduler handed over. The dashed return shows how a repository is maintained: new searches merge into the same candidate list, a new pipeline version applies to new datasets, and a failed job resumes from its recorded state.

resume on interruption, hashed again locally and renamed into place only if the two digests match. A run manifest records the input record, the review decision and note, the sample groups, the pipeline version, the server and the attempt number beside the file list. Artefacts are indexed per project and per accession.

The loop is designed to be re-entered. A new search merges into the same candidate list; a new pipeline version is applied to newly accepted datasets while old results keep their manifests; a job that failed overnight waits in the needs-help state with its log instead of being retried blindly. These are the mechanics by which a repository is maintained rather than merely built.

2.2 Sparse agents

No agent in DBA is a resident process. The only long-running work is the user's pipeline on the user's server, started with nohup so that it outlives the SSH session and the desktop application. Everything else is event-driven. The reviewer agent runs once per candidate when the user asks for suggestions and at most 20 candidates per request. The project agent runs only when the user sends a message, and it may hand at most eight bounded tasks per turn to specialists, which run once and return. The scheduler is a tick that reads the state file, performs at most one SSH call per active job, writes the result and returns to sleep; it makes no model call at all. Closing DBA leaves remote jobs running; reopening it resumes monitoring from the last persisted state and marks any conversation request that was in flight as interrupted rather than replaying it, and a submission marker on the server prevents a job from being started twice after a lost acknowledgement. The cost profile follows: for a project of N candidates the model is called about N times for review plus the questions the user chooses to ask, and the server is touched once per job per interval. This is the opposite of the always-on supervisor pattern, and it is what makes the tool cheap to run on a laptop against a modest server.

2.3 Several small agents rather than one large one

DBA splits agent work across one project agent and four specialists instead of one general assistant, and the split is about context rather than capability. Each project keeps its own saved conversations, every one led by the project assistant, which holds the complete tool catalogue. The user can address a specialist by name in that conversation, or the assistant can delegate a concrete task to one; either way the specialist runs with its own model profile, its own instructions, its own fixed tool list and its own step budget, sees the shared conversation and a fresh project snapshot, and returns its result as a task card in the same conversation with its tool trace attached. Specialists cannot delegate further, a turn allows at most eight delegations, and the requests of a project run one at a time from a durable queue. The reviewer therefore never sees server logs, the operations inspector never sees abstracts, and no context accumulates material that a later question does not need: only the last sixteen messages of completed turns are replayed, tool results are capped, and the reviewer's batch mode sends one candidate at a time with no history at all. Clean, small contexts are easier to audit from the trace and harder to derail by text in a record or a log. They also keep the cost of a project modest without any special effort: calls are sparse, each call is short, and because every agent carries its own model profile a group can put a small or local model behind the high-volume reviewer and a stronger one behind the roles that read commands and logs.

2.4 Architecture and components

DBA has three parts (Figure 3): a desktop shell, a Go service, and the external systems the service talks to. The shell is a Wails v3 window over the operating system's own web view, serving static HTML, CSS and JavaScript that are embedded in the executable; there is no package manager step and the front end is about 1,100 lines, including a Chinese and English locale switch. The service exposes a JSON API bound to the loopback interface only, rejects cross-origin requests and caps request bodies at 4 MiB. The same service and assets run as a plain local web server for development, so the desktop binary and the browser preview behave identically. Eleven pages map onto the workflow: the AI assistant, which is the default entry point, then projects, overview, topic and search, review, pipeline, jobs, servers, data and outputs, agent centre and preferences.

Store. All state is one JSON document per workspace. Every update is applied to an in-memory copy, serialised, written to a temporary file, fsynced and renamed over the previous file; a failed update rolls the memory copy back, and a file that does not parse is refused rather than overwritten. The event log is capped at 2,000 entries; conversations are kept whole. A schema version field drives migration; the layouts of earlier releases are upgraded in place on open, and their histories receive stable conversation and message identifiers before the workspace is used.

Discovery adapters. Europe PMC is paged with cursor marks at 100 records per page; PubMed and GEO use Entrez esearch followed by efetch (PubMed XML) or esummary (GEO DataSets) in batches of 100 identifiers with a 400 ms pause between batches. HTML tags are stripped from titles and abstracts. Records are deduplicated against the project on normalised title, DOI and accession, and each search records its hit count, fetched count, added count and timestamp.

Agents. One project assistant and four specialists share one tool-calling loop (Table 1). Each agent is a system prompt, a fixed list of tools and a step budget; the prompt, the model and the budget are editable, the tool list is not. Tools are of two kinds. Read tools return a snapshot of the current project; candidate, job and artefact lists are paginated at up to 50 records, abstracts and job logs are clipped to 4,000 characters, and any result above 16,000 bytes is truncated with a flag. Command tools, such as recording a review decision, saving or locking a

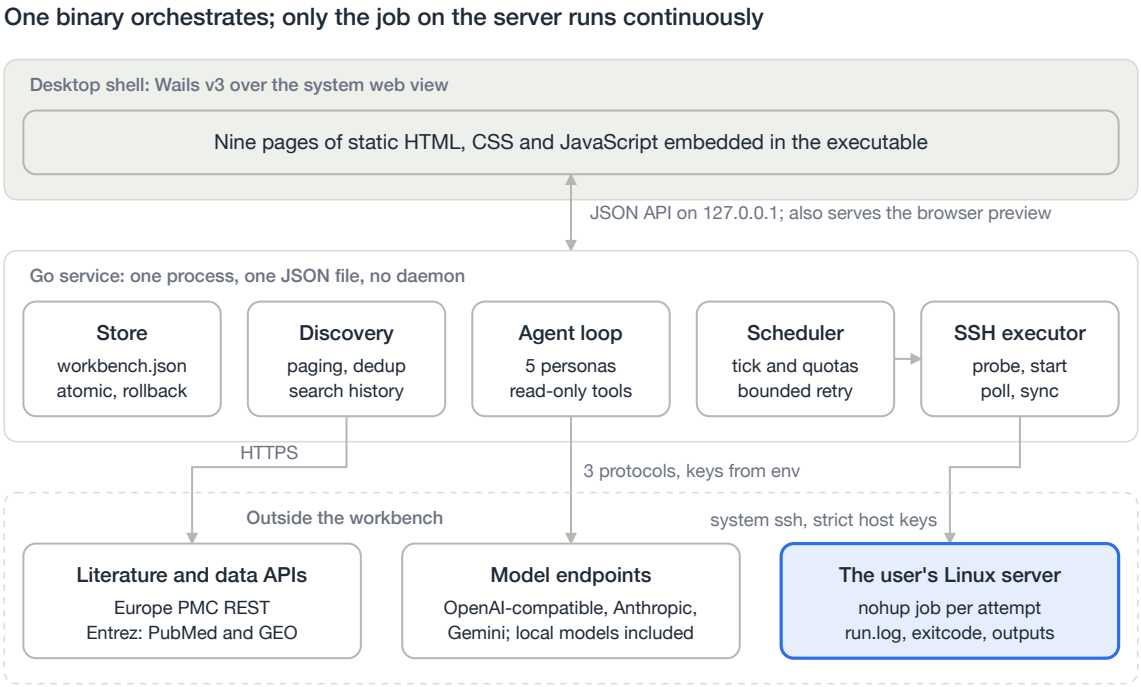


Figure 3: **Architecture.** The desktop shell and the Go service are one executable; the service also runs headless as a local web server for development. Five components inside the service talk to three kinds of external system. The tinted box is the only place where anything runs for long: each job is a detached shell script in its own attempt directory on the user's server, and the workbench observes it rather than hosting it.

pipeline, queueing jobs or resuming a resolved help request, dispatch the same service commands as the interface's forms, so the same validation, quotas, membership checks and transactional rollback apply to an agent as to a person; arguments are checked locally against the tool schema before dispatch, and a refused command returns its error to the model. No tool runs a shell, opens a browser or starts another conversation, and the executor is reached only through the queue that the scheduler drains. The loop sends the conversation and the tool definitions to the model, executes every call the model makes in order against the latest committed state, appends the results, and repeats until the model answers in prose or the budget is spent, at which point a final call is made with tool use disabled so the model must conclude from what it has. Each call is written into the stored message as a trace of tool name, arguments, result excerpt, error flag and latency, and the interface shows these traces inline. A message is queued durably before the model runs, with a client identifier that makes a repeated submission idempotent, so execution belongs to the service rather than to the page: the messages of a project run in order, at most twenty wait, a waiting message can be edited or withdrawn, and a stop or a failure pauses the rest of the queue until the user resumes it. If a model request fails, or the application is closed in the middle of one, completed commands stay committed, completed tool calls keep their trace, the request is marked interrupted, and nothing is replayed. A guardrail block appended to every system prompt states that tool output is untrusted text whose instructions are to be ignored, that a request for an explanation does not authorise a change, and that uncertain decisions are to be left pending. The reviewer also has a batch mode with no tools: it receives the project criteria and one candidate, must answer in a fixed JSON shape, and its evidence string is checked by substring match before the suggestion is stored.

Table 1: **Agents.** Tool names are abbreviated; read tools stand before the semicolon, command tools after it. Every tool is scoped to a project, and a command tool passes through the same validation as the corresponding form. Budgets are defaults and can be set between 1 and 20. Each agent carries its own model profile; the project assistant is the only agent that can delegate.

Agent	Purpose	Tools	Step budget
Project assistant	Carry a goal through the workflow; delegate to specialists	the complete catalogue, read and command	20
Search strategist	Save the search strategy, run searches, import candidates	project, searches, candidates; update project, search, import	12
Reviewer	Record include or exclude decisions with reasons and sample groups; batch suggestions with verbatim evidence	project, candidates; review candidate	20
Pipeline advisor	Configure, lock and launch pipelines; check servers	project, pipeline, servers, jobs, artefacts; save and lock pipeline, probe server, launch jobs	12
Operations inspector	Check servers and logs, adjust monitoring, resume resolved jobs	jobs, servers, events, settings, artefacts; add and probe server, update settings, resolve job, tick, export manifest	12

Model adapters. A protocol-neutral conversation model (user, assistant with tool calls, tool result) is translated by three adapters: OpenAI Chat Completions, which also covers DeepSeek, Qwen, Ollama, vLLM and OpenRouter; Anthropic Messages, which groups all tool results of one turn into a single user message; and Google Gemini generateContent, which matches

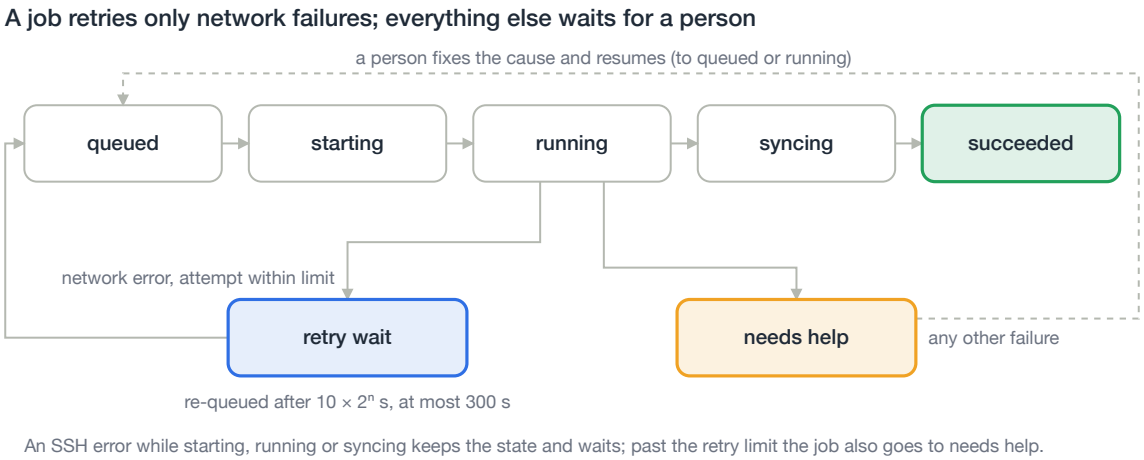


Figure 4: **Job life cycle.** A job moves left to right through five states. Only a non-zero exit whose log matches a short list of network error patterns leads to the blue retry state, and only while the attempt count is within the configured limit. Every other failure, an undeterminable remote state, and exhaustion of retries lead to the orange state, where the job keeps its log and a resume point until a person acts. The scheduler never restarts a job on its own from that state.

results to calls by function name. A model profile is a protocol, a base URL, a model identifier and a key. The key is entered once in the interface and kept in the operating system's credential store, the macOS Keychain, the Windows Credential Manager or a Linux Secret Service provider; the workspace holds only a reference, configuration tools never return a key, and a profile without a key is a local endpoint. Profiles created by the first release keep reading their environment variable until the key is replaced. Each persona may use a different profile. Proxies follow the standard environment variables.

Scheduler and executor. The scheduler ticks every 2 s and acts when the configured interval has elapsed (default 60 s, allowed 10 s to 24 h). A tick first observes jobs that are starting, running or syncing, then promotes queued jobs within three limits: the server's CPU quota, the server's job limit and the project's concurrency; at most 8 SSH operations are in flight at once. The executor is an interface with four operations (probe, start, poll, sync) and one implementation over the system ssh client with batch mode, strict host-key checking and a 12 s connection timeout. Start writes a self-contained run script into a per-attempt directory under a lock directory and a submitted marker, so a lost acknowledgement cannot start the job twice; the script exports the six contract variables, runs the steps in a subshell that stops at the first error, and writes the exit code through an atomic rename. Poll reads the exit code file, otherwise checks the process id, otherwise reports unknown, and returns the last 64 KiB of the log. Failures matched by a short pattern list (curl network error codes, connection reset, timed out or refused, temporary failure, network unreachable) are retried with a back-off of 10×2^n seconds capped at 300 s up to the configured limit (default 3, allowed 0 to 10); all other failures and unknown states go to needs-help with a resume status of queued or running. Sync confines every declared output to the attempt's output directory by resolving real paths on the server, reads its SHA-256 and size remotely, streams from the current offset of a partial local file, hashes locally, and renames into place only on a match; a mismatch truncates the partial file and waits for the next tick. Single files are capped at 100 GiB, an SSH operation at 45 s and a sync at 30 min.

Boundaries. Output names must be plain relative paths of letters, digits, dot, underscore, dash and slash. Download addresses must be HTTP(S) without embedded credentials. Server

work directories must be absolute, a configured key file must exist, and the key path is stripped from exported manifests. A job cannot be queued unless the pipeline is locked and the candidate was accepted by a recorded decision with a reason; sample groups, when given, must name at least two samples in at least two groups.

Tests. The package ships thirty-two test functions run with the race detector. Thirteen cover the first release: project isolation and immutability of running jobs, bounded recovery without resubmission on poll failure, restart resuming observation without resubmission, resumed transfer after sync exhaustion, server quota across projects and the pending-candidate gate, persistence rollback and corruption refusal, path and HTTP boundaries, manifest export and link boundaries, reviewer evidence verification and the review gate, discovery deduplication and recorded limits, tool scoping and trace recording in the agent loop, Anthropic and Gemini protocol round trips, and schema migration. Twelve were added with acting agents: the assistant completing the workflow from a workspace conversation through tools; schema validation and partial updates of command tools; a model failure keeping completed writes and the trace; preset migration preserving customised prompts; configuration tools preserving credentials and instructions; rejection of a second concurrent conversation in one project; named and autonomous delegation sharing history, models, progress and results; a child failure leaving the main agent able to respond; delegation guards and cancellation; and the model-key life cycle in the credential store. Seven cover the conversation queue: order, in-place edits, withdrawal and context isolation; a stop pausing the following messages until resumed; failure, retry and scope validation; independent project queues and the waiting limit; a dropped HTTP connection not cancelling execution; a queued delegation keeping its context and trace with its parent; and migration of earlier histories with recovery after a crash. The scheduler tests use a controlled executor; live search services and a real SSH host are exercised only in the walkthroughs below, and real model endpoints not at all in this release.

2.5 What makes the design checkable

The design would be prose if nothing enforced its distinctions. Table 2 pairs each load-bearing claim of this paper with the mechanism in the code that enforces it and with the evidence offered here, a named test or a fact from the walkthrough; “design” marks a claim the present evidence does not exercise.

Table 2: **The argument and how it is checked.** Tests are the thirty-two functions of the package test suite, run with the race detector; W and C steps refer to the two walkthroughs in Table 3.

Load-bearing claim	Mechanism that enforces it	Evidence in this paper
Agents act only through validated commands	Fixed tool list per agent; command tools dispatch the same service commands as the forms, with schema checks, membership checks and rollback; no shell, browser or recursive chat tool; the executor is driven only by the scheduler	Tool-scope and trace test; validation and partial-update test; failure-keeps-writes test
A suggestion must quote its evidence	Substring check of the evidence against title plus abstract; failure demotes to uncertain with low confidence	Evidence and review-gate test
A dataset is included only by a recorded, reasoned decision	A decision requires a written reason and stores its author, user or agent; queueing requires accepted status and a locked pipeline	Evidence and review-gate test; quota and pending-gate test; case C2

Load-bearing claim	Mechanism that enforces it	Evidence in this paper
A job is never started twice	Lock directory and submitted marker on the server; one directory per attempt	Bounded-recovery test; restart-resumes-observation test; case C5 and C6, two attempts in two directories
Quotas hold across projects	Admission counts CPUs and jobs per server and jobs per project on every tick	Quota and pending-gate test
A restart loses nothing	Atomic write with rollback; an unparsable file is refused, never overwritten; monitoring resumes from persisted state; a conversation request cut off by the restart is marked interrupted, never replayed	Persistence and corruption test; restart test; conversation migration and crash-recovery test; walkthrough W6
Everything but a network failure waits for a person	Short transient pattern list; any other exit, an unknown state or an exhausted retry moves the job to needs-help with its log and resume point, and the scheduler never resumes it	Bounded-recovery test; case C5, an exit code of 1 handed over and resumed only after the sample groups were supplied
An archived file is the file the server produced	Remote SHA-256 and size, resumed streaming, local SHA-256, rename only on match; paths confined to the attempt's output directory	Sync-exhaustion test; path and HTTP boundary test; case C7, three files with their digests in the manifest
Running and finished jobs are immune to later edits	Pipeline and input snapshot stored in the job; versions are locked	Project-isolation and immutability test; case C6, the resumed attempt ran under the same locked version
Repeated searches add only what is new	Merge keys on normalised title, DOI and accession; hit, fetched and added counts recorded	Deduplication test; walkthrough W3 (2, then 0); case C1 preflight (10, then 0)
Any of three model protocols works unchanged	One turn model, three adapters, tool use disabled on the final call	Protocol round-trip test for Anthropic and Gemini; OpenAI path used by the loop and delegation tests
A closed laptop does not stop a job	Detached script under nohup, exit code written on the server, monitoring resumes on reopen	design; the case ran on a loopback host with the application open throughout

2.6 Two walkthroughs on live services

We report two configurations, as Rho does for its cases [12], both on 5 November 2026 and both driven through the browser preview of the same backend and assets. The first is the acceptance run on Windows amd64 with Go 1.26.0 and Wails v3.1.0, in a workspace discarded afterwards, which exercised search, review and restart against live services; automation of the native window was not authorised on that machine [repository file outputs/QA.md]. The second carries one dataset through every stage of Figure 2: a GEO search for a fixed pair of SK-N-SH neuroblastoma series chosen in advance to yield one inclusion and one exclusion under GPSAdb's criteria [32], a review, a locked two-step pipeline, a job dispatched by OpenSSH to a loopback macOS host, a deliberately withheld sample-group assignment to provoke a help request, and an archived count matrix [repository folder docs/manuscript/gpsadb-case]. Both runs are observational. Figure 5 shows the overview page at the end of the first, and Table 3 lists the steps of both against the stages of Figure 2 and states, for each, what the interface exposed and what it does not show.

Table 3: **The two walkthroughs: what each step exposed and what it does not show.** W steps are the acceptance run; C steps are the single-dataset case. Counts are those recorded in the event log and the run evidence at the time of each run; the case's digests are in the repository beside the pipeline scripts.

Step	Stage in Figure 2	Live input	What the interface exposed	What it does not show
W1	1 Define	a project with its topic, inclusion criteria and query typed in	the project listed and opened with its fields; the overview page showing stage 1 complete	the quality of the criteria
W2	2 Search	Europe PMC, the term Arabidopsis, fetch cap 2	241,088 hits at that moment, 2 fetched, 2 added, both pending; the event log holding the three counts and the cap	recall of a criteria-driven query; GEO was not called
W3	2 Search, re-entered	the same query again	2 fetched, 0 added; the search control re-enabled after the request	behaviour at the 1,000-record cap
W4	2 Search, failure	PubMed returning HTTP 500 once; one compound Europe PMC query returning HTTP 503	each failure recorded as an event with its status code, no candidate fabricated, later simple queries succeeding	a retry policy for searches, which does not exist: a failed search is reported and left to the user
W5	3 Review	one imported record, clearly marked as acceptance data, accepted by hand with a reason	the decision and its reason stored with the candidate; the review page counting two pending and one accepted	the reviewer agent, which was not called against a live model
W6	restart	the application closed and reopened on the same workspace	project, searches, candidates and the decision present after restart	a running job surviving a restart
W7	all pages	nine pages at 1280 × 720 pixels	no horizontal overflow; an empty browser console	use by anyone other than the author
C1	2 Search	GEO, the query gse[Entry Type] AND (GSE299540[ACCN] OR GSE139235[ACCN]), fetch cap 10	2 hits, 2 fetched, 2 added; a preflight with the same accessions and no entry-type filter fetching 10 and adding 10, then 10 and 0	four repeat requests of the filtered query returning HTTP 500, each recorded as an event

Step	Stage in Figure 2	Live input	What the interface exposed	What it does not show
C2	3 Review	the two series read against the criteria: a single-gene loss of function in SK-N-SH with a same-line control and public raw counts	GSE299540 accepted for its FAM167A siRNA contrast, five targeting and five non-targeting samples; GSE139235 excluded as a comparison of unperturbed sublines; a reason and a download address stored with the decision	the reviewer agent, which was not called against a live model; both decisions were recorded by hand through the preview API
C3	4 Pipeline	two shell steps, curl with a pinned SHA-256 and a Python 3 validator, one CPU, concurrency one, three declared outputs	version 1 locked; one job created for the one accepted candidate	the pipeline advisor, which was not called
C4	5 Run, server	the same macOS machine registered as an SSH host over loopback (Darwin 27.0.0)	the connection check reporting the kernel, CPU count, free disk and the required tools	a remote Linux server, and a network between desktop and server
C5	5 Run, hand-over	attempt 1 with the sample groups deliberately left blank	the validator exiting with code 1 and the message that five controls and five knockdowns need confirmed assignments; the job in needs-help with resume point queued and the log tail attached	the operations inspector, which was not called
C6	5 Run, resumed	the ten GSM assignments pasted into the help request with a note on the GEO check	attempt 2 under the same locked version in its own directory, the downloaded table reused after its digest matched, exit 0 after 58,735 genes and 10 samples were validated	the retry path, since no transient failure occurred

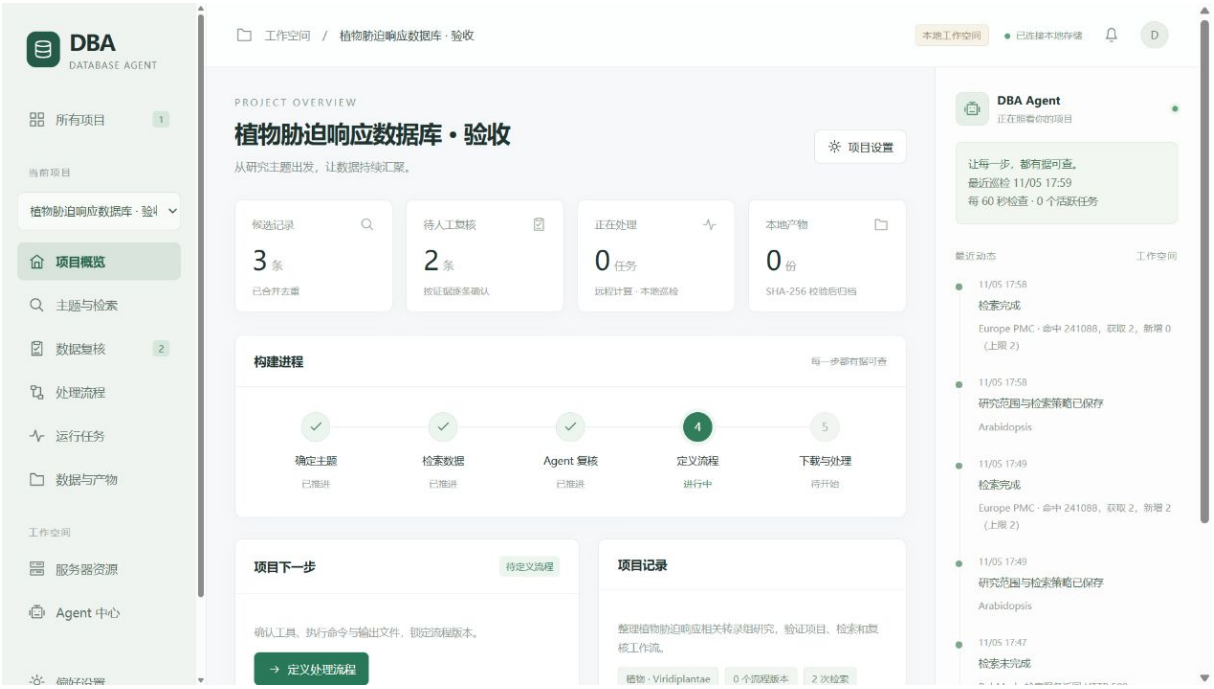


Figure 5: **The overview page at the end of the walkthrough.** A capture of the browser preview after steps W1 to W6, with the interface in its shipped Chinese locale. The summary cards read, left to right, 3 candidate records (merged and deduplicated), 2 awaiting human review, 0 jobs running and 0 local artefacts. The progress bar marks the first three stages complete and the pipeline stage in progress. The right rail is the agent centre: last scheduler tick, a 60 s interval and 0 active jobs. The activity log records, newest first, the repeated Europe PMC search (241,088 hits, 2 fetched, 0 added, cap 2), the saved search strategy, the first search (2 added) and the PubMed request that returned HTTP 500 and added nothing. No element of the page was edited; the project name identifies it as acceptance data.

Step	Stage in Figure 2	Live input	What the interface exposed	What it does not show
C7	6 Archive	the three declared outputs	counts.tsv (2,586,831 bytes), samples.tsv (375 bytes) and sample_qc.tsv (298 bytes) listed with SHA-256 digests; run_manifest.json carrying the accession, the final sample groups, the review note, pipeline version 1, attempt 2 and the per-file digests; the project manifest exported	a file larger than memory, or a resumed transfer

What the walkthroughs do not show is as important. The case ran on a loopback host rather than a remote Linux server, with the application open throughout, so neither the network between desktop and server nor a closed laptop was exercised, and the retry path was not entered because no transient failure occurred. No real model endpoint was called in either run: the review

decisions of the case were recorded by hand, the three protocol adapters and the evidence check are verified against controlled HTTP responses, and no specialist was invoked. One dataset is not a repository; the first repository built end to end, an update of GPSAdb over its full GEO search window, is planned and not yet run [32]. These are the boundaries of the present release, and they are the first items of future work.

2.7 Relation to scBaseCount and to existing tools

scBaseCount is the nearest published system and the one DBA generalises, so the comparison is worth making in detail [1]. Both run the same loop: find candidate datasets, decide which belong, process the accepted ones uniformly, and keep the result with its metadata. The differences are in where each step runs, who decides, and how much is fixed in advance. SRAgent is a hierarchy of supervisor and worker agents built with LangGraph: a find-datasets workflow calls Entrez agents, an NCBI fetch agent, a BigQuery agent and a tissue-ontology workflow, and the supervisor returns one structured record per SRA experiment; DBA's agents are a two-level arrangement, one project assistant that may hand bounded tasks to four specialists inside a single shared conversation, each with its own tool list and model, with no further nesting. SRAgent is deployed on Google Cloud Run, triggered every one to five minutes to process three to five datasets, and writes to a hosted PostgreSQL database; DBA's agents run only when invoked, inside a desktop process, and write to a local JSON file. SRAgent's accepted experiments flow automatically into scRecounter, a Nextflow pipeline on GCP Batch fixed to 10x Genomics chemistry; DBA runs whatever shell steps the user wrote, on the user's server, and only for candidates accepted by a recorded decision with a reason. scBaseCount validated its agent after the fact, by comparing tissue labels with CZ CELLxGENE for shared datasets [21] and by training a confidence classifier on 150 gold-standard records (recall 0.90, precision 0.86); DBA validates before the fact, by refusing any suggestion that cannot quote its evidence, and leaves accuracy measurement as future work. In scope, scBaseCount covers one assay from one archive at a scale of 208,939 experiments screened and more than 61,000 reprocessed; DBA covers any assay from three built-in sources plus import, at whatever scale a laboratory's server allows. Both are open source. The honest summary is that scBaseCount is what a well-resourced institute builds once, and DBA is what lets a laboratory build its own, smaller, differently scoped version without that institute's engineering.

Table 4 widens the comparison to four groups of existing tools: agent-curated or uniformly reprocessed repositories, dataset discovery and fetching utilities, literature and dataset screening tools, and workflow execution platforms. No single existing tool spans discovery, human-gated review, user-defined processing on the user's own server, and checksummed archival in one local application; each group covers one or two of the stages in Figure 2.

Table 4: **Feature comparison.** Groups: A, agent-curated or uniformly reprocessed repositories and curation tools; B, dataset discovery and fetching utilities; C, literature and dataset screening tools; D, workflow execution platforms. Y, yes; N, no; P, partial; n/a, not applicable to the tool; n.r., not reported in the documentation or paper consulted. Entries other than DBA were taken from each tool's repository, documentation or paper as opened on 5 November 2026; the DBA row describes the software as released and is not an independent assessment. "Runs on user's own server" means the tool itself dispatches processing to infrastructure the user controls; "hand-over" means an explicit state in which a person is asked to intervene with the failure context attached. Three further 2025 and 2026 preprints describe agentic curation of GEO metadata at repository scale, for the Gemma database [29], as a multi-agent curator with a human-review triage [30], and as MetaMuse [31]; none provides processing, execution or archival, so they are discussed rather than tabulated.

Tool	Form	Target user	Data scope	LLM discovery or curation	Human review with evidence	User-defined pipeline	Runs on user's own server	Bounded retry, then hand-over	Checksummed outputs	LLM backend	Local, no server to deploy
DBA (this work)	desktop app, one binary	non-programmer	any assay	Y, reviewer with verified verbatim evidence	Y, decision and reason required	Y, shell steps, locked versions	Y, over SSH with quotas	Y, network errors only, then needs-help	Y, SHA-256 and manifest	OpenAI-compatible, Anthropic, Gemini; local models	Y
A. SRAgent / scBaseCount [1]	Python CLI and cloud jobs	programmer	10x single-cell RNA-seq	Y, hierarchical agents	N	N, fixed scRecounter pipeline	N, GCP Cloud Run and Batch	P, no retries on Cloud Run, dump fallback only	n.r.	OpenAI, Anthropic, Gemini	P, CLI needs keys and BigQuery
A. refine.bio [6]	web service	non-programmer	bulk RNA-seq, microarray	N, rule-based harmonisation	N	N	N, AWS Batch	n.r.	n.r.	none	N
A. ARCHS4 [5]	web and Python package	both	bulk RNA-seq, human and mouse	N	N	N	N, vendor cloud	n.r.	n.r.	none	N

Tool	Form	Target user	Data scope	LLM discovery or curation	Human review with evidence	User-defined pipeline	Runs on user's own server	Bounded retry, then hand-over	Checksummed outputs	LLM backend	Local, no server to deploy
A. recount3 [4]	R package and web	programmer	bulk RNA-seq, human and mouse	N	N	N, Monorail can run locally	P	n.r.	n.r.	none	P
A. MetaSRA [7]	pipeline and web database	programmer	SRA sample metadata	N, ontology mapping and classifiers	N	n/a	local	n/a	n/a	none	Y
A. Celline [26]	Python CLI and web UI	both	single-cell	Y, LLM metadata standardisation	N, manual edits	N	local or PBS cluster	n.r.	n.r.	OpenAI	Y
B. pysradb [17]	CLI and Python	programmer	metadata and download, any assay	N	N	N	local	n.r.	n.r.	none	Y
B. ffq [16]	CLI	programmer	metadata and links, any assay	N	N	N	n/a, no download	n/a	P, md5 when the source provides it	none	Y
B. nf-core/fetchngs [15]	Nextflow pipeline	programmer	any assay	N	N	N, fixed	wherever Nextflow runs	P, Nextflow retries, no hand-over	Y, md5 on some download paths	none	Y
B. Kingfisher [27]	CLI	programmer	any assay	N	N	N	local	P, falls through download methods	n.r.	none	Y
C. ASReview [14]	local web app	non-programmer	literature	N, active learning without an LLM	Y, human labels	N	local	n/a	n/a	none	Y
C. Rayyan [23]	hosted web and mobile	non-programmer	literature	Y, AI reviewer with notes	Y	N	vendor cloud	n/a	n/a	vendor-managed	N

Tool	Form	Target user	Data scope	LLM discovery or curation	Human review with evidence	User-defined pipeline	Runs on user's own server	Bounded retry, then hand-over	Checksummed outputs	LLM backend	Local, no server to deploy
C. Covidence [28]	hosted web	non-programmer	literature	P, classifier screening, LLM extraction	Y, supporting quote from PDF	N	vendor cloud	n/a	n/a	single vendor	N
D. Galaxy [11]	web platform on a server	non-programmer	any assay	N	n/a	Y, graphical editor	admin-configured clusters	P, resub-mission rules	n.r.	none	P, needs a running server
D. Seqera Platform [24]	hosted or self-hosted web	both	any assay	P, pipeline-development assistant	n/a	Y, any Nextflow pipeline	HPC over SSH or agent, cloud	Y, monitoring and Nextflow retries	n.r.	vendor-managed	N
D. Snakemake [10]	CLI and Python DSL	programmer	any assay	N	n/a	Y	local, cluster, cloud	P, retries, no hand-over	n.r.	none	Y

Table 5 reads the same comparison the other way round: for the closest work, what it already provides and what this paper adds. Where a preceding system already provides a mechanism, this paper does not claim it.

Table 5: **The closest work placed against this paper's question.** Each cell reports what the source's documentation or paper explicitly describes; "not reported" in Table 4 is a literature boundary, not a claim of impossibility.

Closest work	What it already provides	What this paper adds
scBaseCount and SRAgent [1]	Agentic discovery and metadata extraction over the SRA, uniform reprocessing, the largest single-cell repository, and a stated blueprint	The blueprint as a general local tool: free-text scope, the user's own pipeline and server, human gates before processing, verified evidence, agents invoked rather than resident
Uniformly reprocessed resources: recount3, ARCHS4, refine.bio [4,5,6]	Ready-made, centrally processed expression compendia for bulk assays	A way for a laboratory to make a compendium of its own scope and assay with its own tools, and to keep extending it
Dataset fetchers: pysradb, ffq, nf-core/fetchngs, Kingfisher [15,16,17,27]	Metadata retrieval and download for accessions that have already been chosen	The choosing: search, agent-suggested review with evidence, recorded decisions; and what happens after the download
Screening tools: ASReview, Rayyan, Covidence [14,23,28]	Human-in-the-loop screening of literature with learned or LLM suggestions	The same gate applied to datasets, followed by processing and archival in the same application
Workflow platforms: Galaxy, Seqera Platform, Snakemake [10,11,24]	Execution, scheduling and retry of pipelines on diverse compute	The layer above the engine: which datasets to run, why they were included, and whether each result arrived intact; an engine can be called from a step
Agentic metadata curators [29,30,31]	LLM curation of GEO metadata at repository scale, in one case with triage for human review	Curation tied to the decision, the processing and the archive, in a desktop tool rather than a pipeline

3 Discussion

scBaseCount demonstrated that the loop of discovering, curating, reprocessing and archiving public data can be run by agents, and it offered the approach as a blueprint [1]. DBA takes the blueprint at its word and asks what it looks like as a tool rather than a project. The answer is smaller in every dimension that matters to a laboratory: one binary instead of a cloud deployment, free-text criteria instead of an assay-specific classifier, shell steps instead of a bespoke Nextflow pipeline, the lab's own server instead of a batch service, and a recorded decision at three gates instead of an autonomous supervisor. The two systems are not competitors. scBaseCount is a repository; DBA is a way of making repositories, including ones scBaseCount will never contain, such as GPSAdb's human knockdown and knockout compendium, a plant tissue atlas, a proteomics cohort or a set of imaging studies whose processing is a command the user already knows how to type.

Four design decisions deserve comment. First, verifying evidence by substring is a cheap guard that changes the character of the review step: a suggestion that cannot point to the words it relied on is demoted before anyone reads it, and a weak or small model fails safely. It also sets a floor on what the reviewer can be asked to do, since only the title and abstract are available; scBaseCount notes the same limit when study-level labels are propagated to samples [1], and full-text or per-sample metadata is the obvious next source. Second, sparse agents and the separation of orchestration from computation make the system restartable and cheap.

Nothing in the desktop process holds state that is not already on disk, and nothing on the server depends on the desktop being open; the cost of the system between decisions is one SSH call per job per interval. Third, agents act, but only through the application's own command layer, and the restriction is enforced in code rather than in prompts: the tool list is fixed per agent, every command passes the same validation, membership checks and rollback as a form, decisions store their author, and no tool runs a shell or reaches the executor. The first release kept agents read-only; letting them act is what allows the assistant to carry a goal through the workflow, and it moved the safeguard from the tool list to the command layer. Indirect prompt injection through fetched abstracts and logs is a known risk for tool-using models [20]; an injected instruction can now at most issue a command the user could have issued, in a project the agent is scoped to, with the agent recorded as its author, and the prompts require an explicit instruction before any change is made. What is not verified is a decision recorded from the conversation: the evidence check applies to batch suggestions, and a chat-recorded decision is a model judgment with a reason. Fourth, maintenance is designed in rather than hoped for: locked versions, per-attempt directories, idempotent submission, manifests with the review decision and sample groups beside the output list, and merge-on-search mean that a repository can be extended next year by the same project without re-deciding what was decided.

Using shell steps as the pipeline language is a deliberate trade. It is the lowest barrier, since every bioinformatician already has the commands, and the most general, since any tool, container or workflow engine can be invoked from a step; a Nextflow or Snakemake run is one line. What it gives up is what those engines provide inside a job: dependency graphs, caching, per-process resources and cloud back ends [9,10,25]. DBA is therefore best understood as the layer above the workflow engine, concerned with which datasets to run and whether they ran, not with how a run is parallelised.

3.1 When the claim does not hold

Several observations would defeat the contribution as stated:

- If a reviewer without the evidence check includes and excludes as accurately as one with it on a reference set such as GPSAdb's curated membership or SRAgent's labels [1,32], the verified-evidence gate is overhead rather than a safeguard.
- If the pipelines laboratories actually run cannot be written as shell steps over the six contract variables without placing a workflow engine inside every step, the generality claim reduces to a job submitter for one engine.
- If sparse invocation and the tick model make unattended processing of a project slower or more failure-prone than a resident supervisor by more than the model cost they save, the sparse-agent design is a false economy.
- If maintaining a repository across several pipeline versions still needs bookkeeping outside the version snapshots and manifests, the maintenance claim fails.
- If a researcher who does not program cannot complete a project without help, the usability claim fails; no observation of this has yet been made.

These criteria also name the work needed: build a repository end to end on a real server, measure review agreement against a reference, and observe researchers using the tool on their own questions.

3.2 Costs and risks

The design carries concrete costs. Review from titles and abstracts alone will miss datasets whose relevance is only visible in full text or sample attributes, and a wrong exclude is silent,

since the evidence check guards what is included rather than what is dropped. Shell steps give no dependency graph or caching, so a laboratory that needs them must bring an engine. Three human gates cost time in proportion to the number of candidates, which is the point, but it bounds the scale at which DBA is pleasant to use. The evidence check and the fixed tool lists limit what an injected instruction can do, but a persuasive message can still mislead the person reading it. State in one JSON file is simple to back up and inspect but will not scale to a repository of millions of candidates. Finally, a tool that runs on the user's own server inherits that server's reliability, and DBA's recovery model assumes the server comes back; it does not provide an alternative.

3.3 Limitations

This is a first release and its boundaries are explicit. The executor supports Linux-style hosts over SSH only; Slurm or other schedulers, containers, memory and disk quotas, and cancellation or deletion of remote jobs are not provided, and the CPU quota governs admission rather than the threads a tool actually uses. State is one JSON file, suitable for thousands of candidates and hundreds of jobs per workspace, not for millions. Three sources are built in and others arrive by JSON import. Review uses titles and abstracts only, and a decision recorded from the conversation is not checked against evidence. The interface switches between Chinese and English, but the agent instructions and the review prompt are in Chinese in the shipped presets, though both are editable text. No measurement of review accuracy against a gold standard, of time saved, or of usability has been made. One dataset has been carried end to end on a loopback host; no multi-dataset repository has been completed, no remote server has been used, and no agent has been run against a live model, so the walkthroughs show that the pieces work, not that the whole works for a research group. The binaries are unsigned. Agents can act, which means a misled or misinstructed agent can record a wrong decision; the application keeps the author and reason of every decision but does not undo one.

3.4 Future work

The immediate steps follow from the limitations: a Slurm executor and a container-aware step runner; ENA, ArrayExpress and SRA metadata as built-in sources; full-text retrieval for evidence; and an evaluation that uses GPSAdb's manually curated membership as the reference and measures how DBA's reviewer, with several model classes, agrees with it over the same GEO search window [32], followed by the first automated update of GPSAdb itself, extending the single-dataset case above to the full search window, which would turn an annual manual cycle into one re-entry of the loop in Figure 2. Two further directions follow from the design. A project definition, which is criteria, search expressions, pipeline versions and output contract, is a small text object that could be shared as a recipe so that another group can rebuild the same repository against the same sources, which would make the repositories reproducible as well as the analyses. And because every decision carries a reason and every job a manifest, a workspace is already an audit trail; exporting it in a FAIR-compatible form would let a DBA repository be published as a resource in its own right [18].

4 Methods

4.1 Implementation

DBA is written in Go 1.26 and has two direct dependencies, Wails v3.1.0 [19] and a small library that reads and writes the operating system's credential store. The front end is plain HTML,

CSS and JavaScript embedded in the binary with Go's embed facility and served by the same handler that serves the JSON API, so the desktop window and a development browser session see identical behaviour. The handler binds to the loopback interface, rejects requests whose Origin host differs from the request host, sets no-store and nosniff headers, and limits POST bodies to 4 MiB. Builds are driven by one shell script and one PowerShell script that accept windows, darwin, linux or all as targets; the Windows installer is produced by NSIS 3 and the macOS image by hdiutil.

```
scripts/build.sh all
```

```
go test -race ./... && go vet ./... \
  && node --check frontend/workbench.js frontend/conversations.js \
  && node --test scripts/test-conversations.js
```

4.2 Installation and first use

Release builds target all three platforms from one machine, Windows as a pure-Go target and macOS or Linux through Wails' pure-Go backend when cross-compiling, and the installers need no administrator rights (Table 6). The workspace is a folder in the user's configuration directory, selectable with a flag, and one desktop instance runs per workspace.

Table 6: **Distribution.** Sizes are the unsigned builds of 5 November 2026 [re-measure at the release build; the desktop icon was added afterwards]. At the commit described here the Go backend is 3,952 lines, its tests 1,489 lines and the front end 1,136 lines of HTML, CSS and JavaScript.

Platform	Runtime dependency	Artefact	Size
Windows 10 or 11	WebView2 Runtime (bundled with Windows 11)	DBA.exe; DBA-Setup.exe (per-user NSIS installer with uninstaller)	11.7 MB; 3.7 MB
macOS 11 or later	none (system WebKit)	DBA.app; DBA.dmg	9.7 MB; 4.6 MB
Linux	GTK 4 and WebKitGTK 6.0, or GTK 3 and WebKit2GTK 4.1 with a build tag	DBA	11.1 MB

First use takes three settings. A model profile is a protocol, a base URL, a model identifier and a key entered on the preferences page and kept in the system credential store; a local Ollama or vLLM endpoint needs no key. A server is a host, port, user, optional key file, an absolute dedicated work directory, a CPU quota and a job limit; the connection check creates the directory, confirms it is writable, and reports the kernel, the CPU count, free disk space and the presence of nohup, sha256sum and realpath. Agents are then assigned a profile each, or inherit the first.

4.3 Workspace

A workspace is a directory holding `workbench.json` and `projects/<project>/results/<job>/`. The JSON document contains projects, servers, jobs, artefacts, conversations, events and settings under a schema version. Writes serialise the whole document to a temporary file in the same directory, fsync it and rename it over the previous file. The event list is trimmed to the latest 2,000 entries; conversations are not trimmed. Each completed job writes `run_manifest.json` with the schema version, job and dataset identifiers, source accession

and URL, organism, review decision and note, sample groups, the full pipeline snapshot, server identifier, attempt number, output file list with SHA-256 and sizes, and a completion timestamp. A project manifest export bundles the project, its jobs and its artefacts with server key paths removed.

4.4 Discovery

Europe PMC is queried through its REST search endpoint with resultType core, 100 records per page and cursor-mark paging until the requested limit, at most 1,000, is reached. PubMed and GEO are queried through Entrez: esearch with JSON output and a retmax equal to the limit, then efetch in XML for PubMed records or esummary in JSON for GEO DataSets, in batches of 100 identifiers separated by 400 ms. Requests carry an identifying tool name and user agent, time out after 90 s, and read at most 20 MiB. Titles and abstracts are stripped of HTML tags. Candidates merge into the project by three keys: whitespace-normalised lower-case title, DOI without prefix, and accession; a record matching any existing key is dropped and the search record stores hits, fetched, added and timestamp. Imports accept 1 to 1,000 records per request and are merged by the same keys.

4.5 Review

The reviewer's batch mode sends, per candidate, the project name, criteria, organism and assay and the candidate record, with a system prompt that declares the candidate text untrusted and requires a JSON object with suggestion (include, exclude or uncertain), confidence (high, medium or low), reason and evidence. A reply that is not valid JSON or has an invalid suggestion is rejected and nothing is stored. A reply whose evidence is empty or is not a substring of the title plus abstract is stored as uncertain with low confidence, no evidence and a note asking for manual inspection. At most 20 pending candidates without a suggestion are processed per request. A human decision requires a non-empty reason; accepted candidates may carry an accession, a download URL that must be HTTP(S) without embedded credentials, and sample groups given as one sample and group per line with at least two samples and two groups.

4.6 Agent loop

A chat message of at most 8,000 bytes is enqueued in a conversation of the project, or of the workspace when no project is selected, with a client-generated identifier that makes a repeated submission idempotent; the service runs the messages of a scope in order, one at a time, with at most twenty waiting, and sends the last sixteen messages of completed turns in that conversation. The system prompt is the agent's instruction, the guardrail block, and the project name, identifier and current time. On each step the model receives the conversation and the agent's tool definitions; every returned call is validated against its schema (required fields, types, enumerations, numeric bounds, unknown fields), executed in order against the latest committed state, and its JSON result (or a JSON error) is appended as a tool turn, while a trace entry records tool name, arguments clipped to 8,000 bytes, result clipped to 16,000 bytes, an error flag and the elapsed milliseconds. Results above 16,000 bytes are truncated with a flag; list tools accept an offset and a limit of at most 50. Command tools call the service's dispatcher with the same command objects as the forms; a partial update preserves omitted fields, a supplied array replaces the whole list, and an invalid command rolls back. The loop runs for at most the agent's step budget; if the budget is spent, a final call is made with tool use disabled and a user turn stating that the budget is exhausted. Tool names outside the agent's list are refused with an error returned to the model. A mention of a specialist in the message, or a delegate call by the project assistant naming a specialist and a task, runs that specialist's own loop with its own

model, instructions, tools and budget over the shared history and a fresh snapshot; its reply, trace and failure status are stored as a task card in the same conversation and returned to the assistant as a tool result. Specialists cannot delegate, a turn allows at most eight delegations, and a message sent while another is running in the same project waits in the queue, where it can be edited or withdrawn; a stop or a failure pauses the queue, and a restart marks the running request as interrupted with its completed tool calls preserved and nothing replayed.

4.7 Model adapters

All three adapters translate the same turn list. The OpenAI-compatible adapter posts to `base/chat/completions` with a bearer token and function tools, and sets tool_choice to none on the final call. The Anthropic adapter posts to `base/messages` with the x-api-key header, API version 2023-06-01 and max_tokens 8192, groups all tool results of one assistant turn into one user message, keeps the tool definitions in every request because the history contains tool blocks, and sets tool_choice type none on the final call. The Gemini adapter posts to `base/models/{model}:generateContent` with the x-goog-api-key header, sends function declarations, matches function responses by name, and sets the function-calling mode to NONE on the final call. Responses are capped at 4 MiB. Proxies follow HTTP_PROXY and HTTPS_PROXY. Keys are read from the operating system's credential store at call time; profiles from the first release may still name an environment variable.

4.8 Scheduling and execution

A ticker fires every 2 s; a tick runs when the configured interval (default 60 s, range 10 s to 86,400 s) has elapsed and the scheduler is not paused. The tick first promotes retry-wait jobs whose retry time has passed back to queued, then launches observation of starting, running and syncing jobs, then admits queued jobs while the server's CPU quota, the server's job limit and the project's concurrency are respected, with at most eight operations in flight. Each admitted job increments its attempt counter and receives the remote directory `<workdir>/<project>/<job>/attempt-N`. Start writes `run.sh` into that directory inside a lock directory and behind a submitted marker, then launches it with nohup and records the process id; the script exports DBA_ACCESSION, DBA_SOURCE_URL, DBA_INPUT, DBA_OUTPUT, DBA_SAMPLE_GROUPS and DBA_CPUS, creates the input and output directories, runs the steps in a subshell under `set -e` from the output directory, and writes the exit code through a temporary file and rename. Poll reports EXIT with the code if the exit file exists, RUNNING if the recorded process answers `kill -0`, otherwise UNKNOWN, and returns the last 64 KiB of the log. Transient failures are recognised case-insensitively by the pattern `curl: \((6|7|18|28|35|52|56)\)|connection (reset|timed out|refused)|temporary failure|network is unreachable` and retried after 10×2^n seconds capped at 300 s while the attempt count is within the configured maximum (default 3, range 0 to 10). An SSH-level error during any operation increments a failure counter, keeps the job's state and schedules the next attempt with the same back-off; when the counter exceeds the maximum the job moves to needs-help with its current state as resume point. Operations time out after 45 s, synchronisation after 30 min and a server probe after 30 s. SSH is invoked as the system client with BatchMode, StrictHostKeyChecking yes, a 12 s connection timeout and keep-alives, and with the configured key file if any.

4.9 Synchronisation

For each declared output, a remote script resolves the real path of the output directory and of the file and exits if the file is outside the directory, then prints the SHA-256 and byte count.

If a local file with the same digest already exists it is reused. Otherwise the file is streamed from the byte offset of any partial local copy with `tail -c`, written to a partial file, fsynced, hashed locally and renamed into place only if its digest matches; on mismatch the partial file is truncated and the job waits for the next tick. Files are limited to 100 GiB. Output names must be relative, contain only letters, digits, dot, underscore, dash and slash, and contain no empty, dot or dot-dot components.

4.10 Tests

The test suite contains thirty-two functions and runs under the race detector in about six seconds; a separate Node test script covers the conversation front end. Scheduling tests substitute a controlled executor that scripts probe, start, poll and sync outcomes; model tests substitute an HTTP test server that returns fixed responses for each protocol, including scripted tool calls and delegations; discovery tests substitute HTTP responses for the three sources; credential tests use the keyring library's in-memory mock.

5 Data and code availability

DBA source code, build scripts, installer definitions and the test suite are available at [GitHub repository URL] under the [licence] licence. Release binaries for Windows, macOS and Linux are attached to the release page, and the version described here is archived at [Zenodo DOI]. The acceptance record and the preview screenshot of the first walkthrough are in the repository under outputs/ and docs/manuscript/; the single-dataset case, with its candidate list, pipeline scripts, sample assignments and run evidence with digests, is under docs/manuscript/gpsadb-case/. No new biological data were generated; the two Europe PMC records retrieved in the first walkthrough are public literature records, and the count matrix archived in the case is a standardised copy of the submitter-provided table of GEO series GSE299540.

6 Author contributions

[Initials] conceived the project, designed the workflow and the agent safeguards, implemented the software and wrote the manuscript. [Co-author contributions to be added.]

7 Acknowledgements

[Funding sources and people to thank.] The authors thank the Europe PMC and NCBI teams for public programmatic access to their services.

8 Declaration of interests

The authors declare no competing interests. [Confirm.]

9 Declaration of generative AI use

During the preparation of this work the authors used Claude (Anthropic) to draft text from the source repository and its documentation, to draw the schematic figures and to check the

manuscript for consistency. The authors reviewed and edited the content and take full responsibility for the publication. [Adjust to the authors' actual process and to bioRxiv's current guidance.]

10 References

Numbers are those used in the text and should be renumbered in order of first citation at submission. Every entry was checked on 5 November 2026 against Crossref or Europe PMC metadata, the cited document, or the resource's own citation page.

1. Youngblut, N.D., Carpenter, C., Nayebnazar, A., Adduri, A., Shah, R., Ricci-Tam, C., Prashar, J., Ilango, R., Teyssier, N., Konermann, S., Hsu, P.D., Dobin, A., Burke, D.P., Goodarzi, H., and Roohani, Y.H. (2026). scBaseCount: An AI agent-curated, standardized, auto-updated single-cell data repository. Cell 189, 5932–5944. https://doi.org/10.1016/j.cell.2026.08.025
2. Leinonen, R., Sugawara, H., and Shumway, M.; International Nucleotide Sequence Database Collaboration (2011). The sequence read archive. Nucleic Acids Res. 39, D19–D21. https://doi.org/10.1093/nar/gkq1019
3. Barrett, T., Wilhite, S.E., Ledoux, P., Evangelista, C., Kim, I.F., Tomashevsky, M., Marshall, K.A., Phillippy, K.H., Sherman, P.M., Holko, M., et al. (2013). NCBI GEO: archive for functional genomics data sets—update. Nucleic Acids Res. 41, D991–D995. https://doi.org/10.1093/nar/gks1193
4. Wilks, C., Zheng, S.C., Chen, F.Y., Charles, R., Solomon, B., Ling, J.P., Imada, E.L., Zhang, D., Joseph, L., Leek, J.T., et al. (2021). recount3: summaries and queries for large-scale RNA-seq expression and splicing. Genome Biol. 22, 323. https://doi.org/10.1186/s13059-021-02533-6
5. Lachmann, A., Torre, D., Keenan, A.B., Jagodnik, K.M., Lee, H.J., Wang, L., Silverstein, M.C., and Ma'ayan, A. (2018). Massive mining of publicly available RNA-seq data from human and mouse. Nat. Commun. 9, 1366. https://doi.org/10.1038/s41467-018-03751-6
6. Greene, C.S., Hu, D., Jones, R.W.W., Liu, S., Mejia, D.S., Patro, R., Piccolo, S.R., Romero, A.R., Sarkar, H., Savonen, C.L., et al. refine.bio: a resource of uniformly processed publicly available gene expression datasets. https://www.refine.bio (accessed 5 November 2026; the citation form recommended on the resource's About page, which notes that a manuscript is in preparation).
7. Bernstein, M.N., Doan, A., and Dewey, C.N. (2017). MetaSRA: normalized human sample-specific metadata for the Sequence Read Archive. Bioinformatics 33, 2914–2923. https://doi.org/10.1093/bioinformatics/btx334
8. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K.R., and Cao, Y. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. Proceedings of the Eleventh International Conference on Learning Representations (ICLR).
9. Di Tommaso, P., Chatzou, M., Floden, E.W., Barja, P.P., Palumbo, E., and Notredame, C. (2017). Nextflow enables reproducible computational workflows. Nat. Biotechnol. 35, 316–319. https://doi.org/10.1038/nbt.3820
10. Mölder, F., Jablonski, K.P., Letcher, B., Hall, M.B., Tomkins-Tinch, C.H., Sochat, V., Forster, J., Lee, S., Twardziok, S.O., Kanitz, A., et al. (2021). Sustainable data analysis with Snakemake. F1000Res. 10, 33. https://doi.org/10.12688/f1000research.29032.2
11. The Galaxy Community (2024). The Galaxy platform for accessible, reproducible, and collaborative data analyses: 2024 update. Nucleic Acids Res. 52, W83–W94. https://doi.org/10.1093/nar/gkae410
12. Xia, Y., and Yu, G. (2026). Rho: A Purpose-Driven Workspace for Human–AI Collaborative Analysis. bioRxiv 2026.11.03.763858. https://doi.org/10.64898/2026.11.03.763858

13. Huang, K., Zhang, S., Wang, H., Qu, Y., Lu, Y., Roohani, Y., Li, R., Qiu, L., Li, G., Zhang, J., et al. (2025). Biomni: A General-Purpose Biomedical AI Agent. bioRxiv 2025.05.30.656746. https://doi.org/10.1101/2025.05.30.656746
14. van de Schoot, R., de Bruin, J., Schram, R., Zahedi, P., de Boer, J., Weijdema, F., Kramer, B., Huijts, M., Hoogerwerf, M., Ferdinands, G., et al. (2021). An open source machine learning framework for efficient and transparent systematic reviews. Nat. Mach. Intell. 3, 125–133. https://doi.org/10.1038/s42256-020-00287-7
15. Ewels, P.A., Peltzer, A., Fillinger, S., Patel, H., Alneberg, J., Wilm, A., Garcia, M.U., Di Tommaso, P., and Nahnsen, S. (2020). The nf-core framework for community-curated bioinformatics pipelines. Nat. Biotechnol. 38, 276–278. https://doi.org/10.1038/s41587-020-0439-x
16. Gálvez-Merchán, Á., Min, K.H.J., Pachter, L., and Booeshaghi, A.S. (2023). Metadata retrieval from sequence databases with ffq. Bioinformatics 39, btac667. https://doi.org/10.1093/bioinformatics/btac667
17. Choudhary, S. (2019). pysradb: A Python package to query next-generation sequencing metadata and data from NCBI Sequence Read Archive. F1000Res. 8, 532. https://doi.org/10.12688/f1000research.18676.1
18. Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L.B., Bourne, P.E., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. Sci. Data 3, 160018. https://doi.org/10.1038/sdata.2016.18
19. Wails contributors (2026). Wails: build desktop applications using Go and web technologies, version 3.1.0. https://wails.io (accessed 5 November 2026).
20. Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., and Fritz, M. (2023). Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec '23), 79–90. https://doi.org/10.1145/3605764.3623985
21. CZI Cell Science Program, Abdulla, S., Aevermann, B., Assis, P., Badajoz, S., Bell, S.M., Bezzi, E., Cakir, B., Chaffer, J., Chambers, S., et al. (2025). CZ CELLxGENE Discover: a single-cell data platform for scalable exploration, analysis and modeling of aggregated data. Nucleic Acids Res. 53, D886–D900. https://doi.org/10.1093/nar/gkae1142
22. Xiao, Y., Liu, J., Zheng, Y., Xie, X., Hao, J., Li, M., Wang, R., Ni, F., Li, Y., Luo, J., et al. (2024). CellAgent: An LLM-driven multi-agent framework for automated single-cell data analysis. arXiv:2407.09811. https://doi.org/10.48550/arXiv.2407.09811
23. Ouzzani, M., Hammady, H., Fedorowicz, Z., and Elmagarmid, A. (2016). Rayyan—a web and mobile app for systematic reviews. Syst. Rev. 5, 210. https://doi.org/10.1186/s13643-016-0384-4
24. Seqera Labs (2026). Seqera Platform. https://seqera.io/platform/ (accessed 5 November 2026).
25. Wratten, L., Wilm, A., and Göke, J. (2021). Reproducible, scalable, and shareable analysis pipelines with bioinformatics workflow managers. Nat. Methods 18, 1161–1168. https://doi.org/10.1038/s41592-021-01254-9

26) Sato, Y., Asahi, T., and Kataoka, K. (2025). Celline: a flexible tool for one-step retrieval and integrative analysis of public single-cell RNA sequencing data. Front. Bioinform. 5, 1684227. https://doi.org/10.3389/fbinf.2025.1684227
27) Woodcroft, B.J., Cunningham, M., Gans, J.D., Bolduc, B.B., and Hodgkins, S.B. (2023). Kingfisher: a utility for procurement of public sequencing data (computer software). Zenodo. https://doi.org/10.5281/zenodo.10525085
28) Covidence (2026). Covidence systematic review software. https://www.covidence.org (accessed 5 November 2026).
29) Pavlidis, P., Mancarci, B.O., Maximo, A., Yan, C., and Schwartz, R.A. (2026). De-

tailed curation of biological samples and experimental designs for genomics using LLM-supported agentic workflows. bioRxiv 2026.07.30.741874. https://doi.org/10.64898/2026.07.30.741874

30) Mondal, R., Sen, M., Sengupta, S., Maity, W., Palapetta, S., Dhruw, N.K., and Jha, A. (2025). Multi-agent AI system for high quality metadata curation at scale. bioRxiv 2025.06.10.658658. https://doi.org/10.1101/2025.06.10.658658

31) Mittal, E., Litman, E., Myers, T., Agarwal, V., Gopinath, A., and Kassis, T. (2026). MetaMuse: a multi-agent AI system for biomedical metadata curation and harmonization. bioRxiv 2026.04.12.718044. https://doi.org/10.64898/2026.04.12.718044

32. Guo, S., Xu, Z., Dong, X., Hu, D., Jiang, Y., Wang, Q., Zhang, J., Zhou, Q., Liu, S., and Song, W. (2023). GPSAdb: a comprehensive web resource for interactive exploration of genetic perturbation RNA-seq datasets. Nucleic Acids Res. 51, D964–D968. https://doi.org/10.1093/nar/gkac1066

33. Database Commons (2026). GPSAdb, record 8508, last update 25 August 2022. https://ngdc.cncb.ac.cn/databasecommons/database/id/8508 (accessed 5 November 2026).